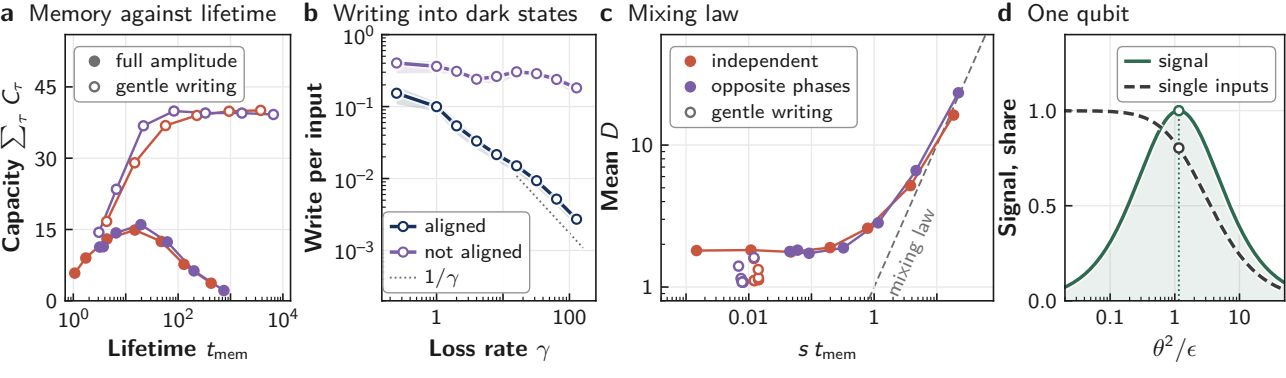

a Memory against lifetime
full amplitude
gentle writing
Capacity $\sum_\tau C_\tau$
45
30
15
0
10^0
10^2
10^4
Lifetime t_{mem}
b Writing into dark states
Write per input
10^0
10^{-1}
10^{-2}
10^{-3}
aligned
not aligned
$1/\gamma$
1
100
Loss rate γ
c Mixing law
independent
opposite phases
gentle writing
Mean D
10
1
mixing law
0.01
1
$s\, t_{\mathrm{mem}}$
d One qubit
signal
single inputs
Signal, share
1.0
0.5
0.0
0.1
1
10
θ^2/ϵ